



Robots Influence in War

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Introduction

THE INFLUENCE THAT ROBOTS CAN HAVE IN WAR IS STILL NEW AND DEVELOPING; HOWEVER, ROBOTS HAVE ALREADY STARTED A SUPPORTING ROLE IN THE MILITARY. THE TYPES OF SUPPORTING ROLES CAN VARY, RANGING FROM MEDICAL SUPPORT, LOGISTICAL SUPPORT, AND PROTECTION ROLES.

Medical Robot To The Rescue!

- ❑ A ROBOT DESIGNED TO REDUCE CASUALTIES BY RESCUING INJURED OR SEVERELY WOUNDED SOLDIERS.
- ❑ THE ROBOT IS CALLED THE BATTLEFIELD EXTRACTION-ASSIST ROBOT - AKA BEAR!

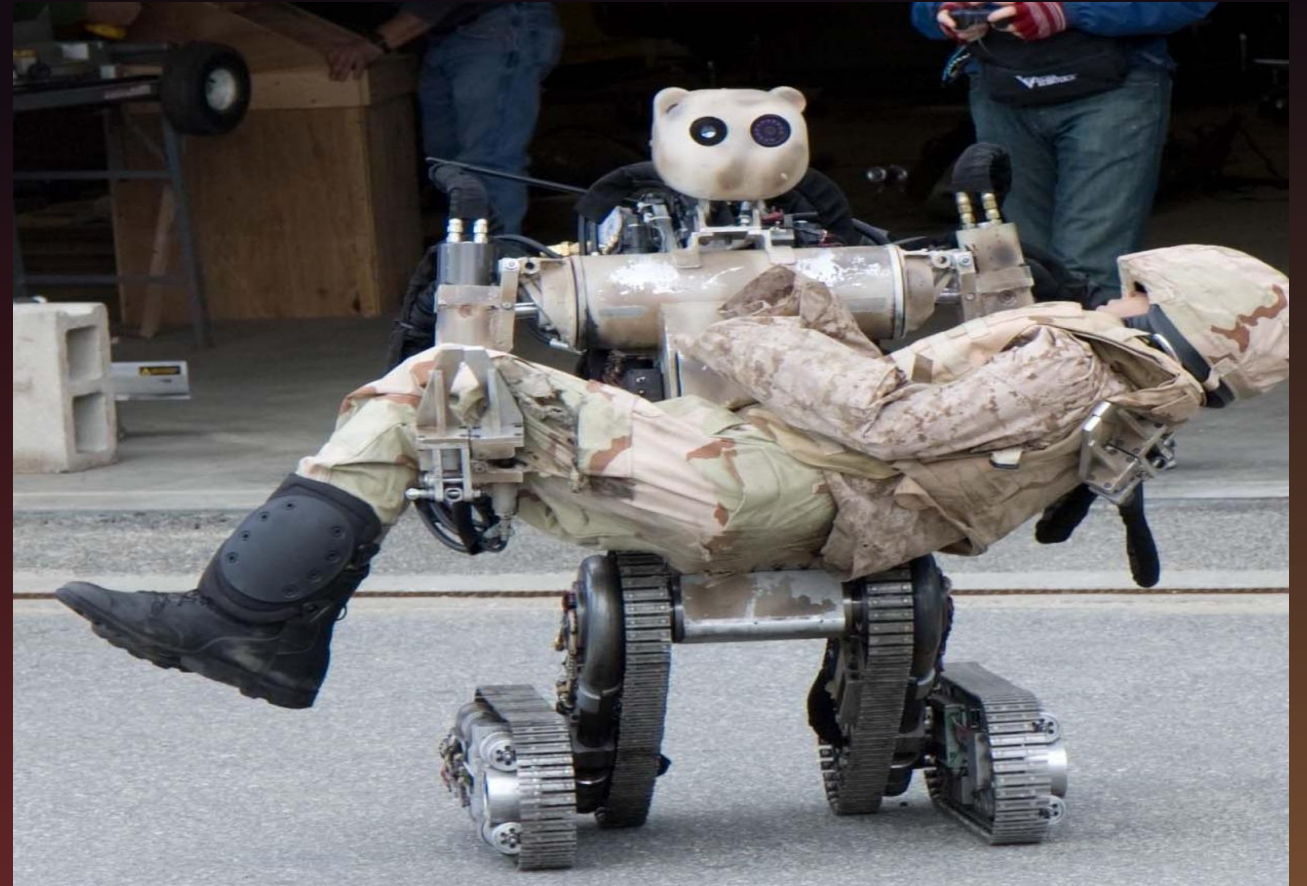


Figure 1: example of what BEAR is designed to do
Source: Adapted from [2]

Medical Robot To The Rescue!

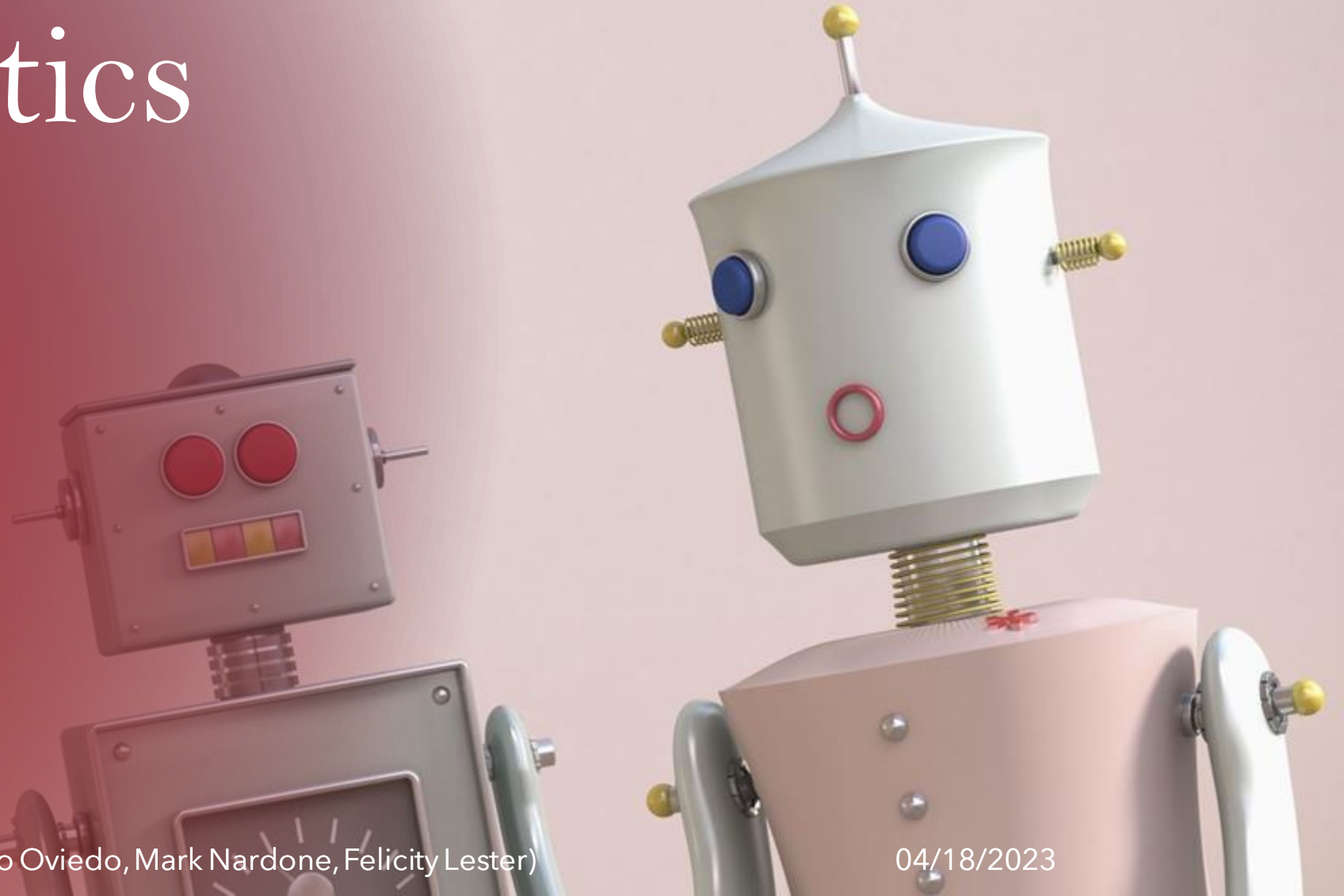


Figure 2: BEAR being remotely controlled
Source: Adapted from [3]

- ❑ THERE IS A GREAT RISK TO RESCUE THOSE WOUNDED WHILE UNDER FIRE.
- ❑ BEAR CAN HELP REDUCE THE RISK BECAUSE IT IS DESIGNED TO CARRY LARGE LOADS OVER A LONG DURATION, AND ON A VARIETY OF TERRAINS [3].

Robots in Logistics

Benefits and examples



Robots in Logistics

- ❑ TITAN UNMANNED GROUND VEHICLE (UGV) CAN AID GREATLY IN THE DELIVERY OF VITAL SUPPLIES AND HUMANITARIAN RELIEF TO THE FRONT LINE [7].
- ❑ CAN HOLD UP TO 2,000 LBS.



Figure 3, 4, 5: Titan Unmanned Ground Vehicle (UGV)
Source: Adapted from [7]



Robots in Protection

- ❑ ROBOTS HAVE BEEN TESTED IN ADDING ADDITIONAL BASE PROTECTION
- ❑ "MILITARY ROBOTS CAN TRAVERSE HAZARDOUS ENVIRONMENTS THAT ARE OTHERWISE FATAL TO HUMANS." [1]
- ❑ SOME SOURCES AVERAGE ABOUT 86% OF BATTLEFIELD DEATHS OCCUR AFTER THE FIRST 30 MINUTES POST-INJURY [5]

Robots in Protection

Figure : A Quad-legged Unmanned Ground Vehicle
Source: Adapted from [4]



Robots in Protection: Quad-legged UGVs

- ❑ ADDS AN ADDITIONAL LEVEL OF BASE PROTECTION [4]
- ❑ GOAL: ALLOW MILITARY PERSONAL TO FOCUS ON MORE CRITICAL AREAS



Figure : First semi-autonomous Quad-legged UGVs
Source: Adapted from [4]

Robots in Protection: MINE CLEARANCE ROBOT

- ❑ ROTATING CHAINED HAMMERS.
- ❑ THE MACHINE DIGS INTO THE SOIL, DETONATION OR SHATTERING ANTI-PERSONNEL MINES [6]
- ❑ SENDS LIVE FEEDBACK TO CREATE A SAFETY RANGE
- ❑ GOAL: REDUCE THE CASUALTIES FROM LAND MINES



Figure : M160 Robotic Mine Flail
Source: Adapted from [6]

Issues and Problems

Figure: Robot and Human
Source: Adapted from [11]



Risk For Mistakes

□ THE USE OF ROBOTS IN WAR WHERE THE ENVIRONMENT IS CHAOTIC PRONES THE RISK OF THE ROBOTS' PROGRAM TO MISINTERPRET SITUATIONS.



Figure: Confused robot
Source: Adapted form [8]

Vulnerable

JUST LIKE A COMPUTER A
ROBOT IS VULNERABLE TO:

❑ HACKING

❑ CYBER ATTACKS



Figure: Hacking image
Source: Adapted from [9]

Human Dependent on Robots

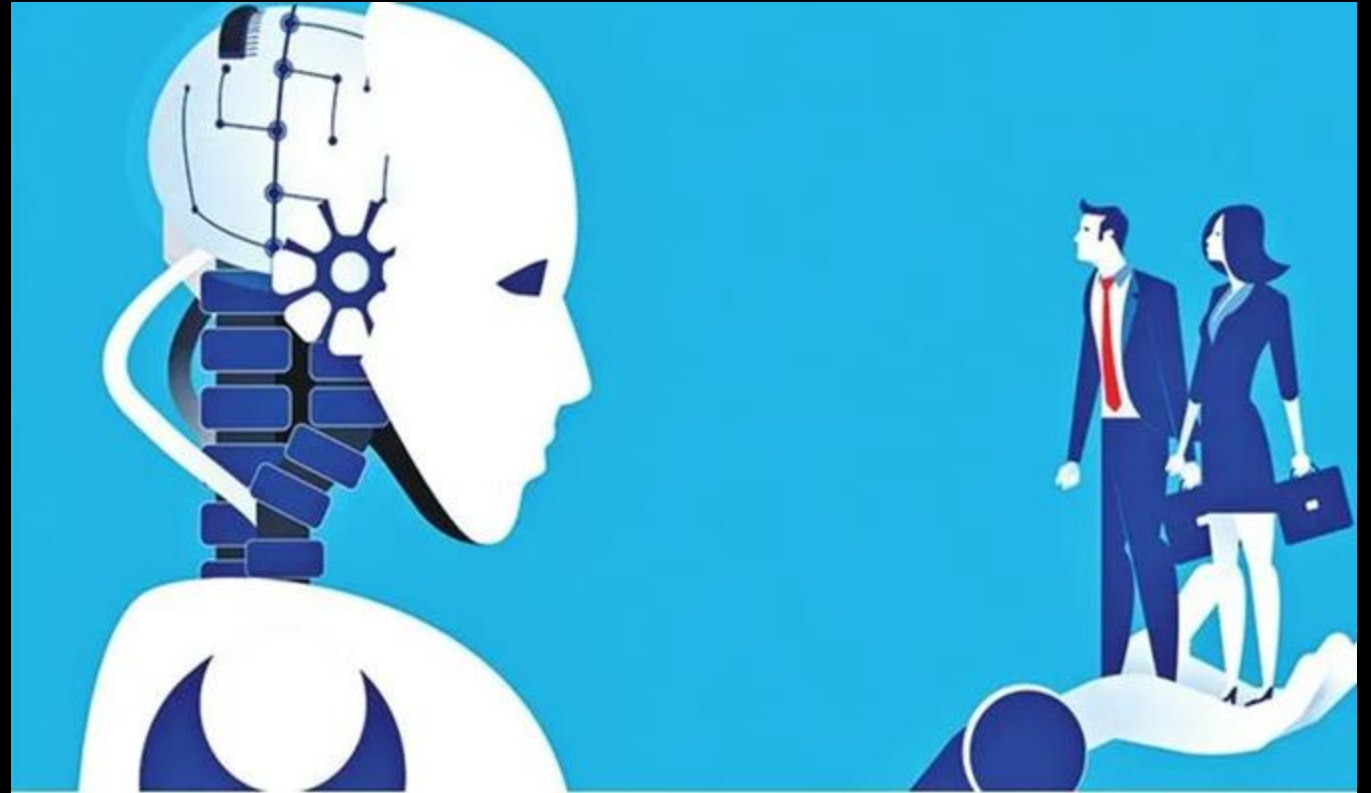


Figure: Robot and Human
Source: Adapted from [10]

THE USE OF ROBOTS CAN LEAD TO
THE MILITARY DEPENDING ON
ROBOTS.

Additional Concerns & Problems

- ❑ TECHNICAL MALFUNCTIONS
- ❑ THE NEED FOR MAINTENANCE AND REPAIR
- ❑ COST
- ❑ JOB LOSS

Analysis of Ethical Issues: Rule Utilitarianism

- ❑ Rule Utilitarianism: Live by rules that are likely to lead to the greatest good for the greatest number.
- ❑ Based on the definition the lack of rules will result in more harm than good.
 - ❑ "What laws or regulations apply, if any? Set by whom?" [2]
- ❑ Undetermined rules can lead to...

Analysis of Ethical Issues: Kants 2nd Formulation

- ☐ Is anyone being used as a means to an end?
- ☐ Robots are being used to protect humans

Analysis of Ethical Issues: Act Utilitarianism

- ❑ Act Utilitarianism:
- ❑ Soldiers' ability to consent to risk
- ❑ In the instance of robot
- ❑ Use of robots is still new and developing - malfunctions do happen
- ❑ Should soldiers be notified of new potential risk?

Summary

- ❑ IN CONCLUSION ROBOTS ARE STILL NEW AND DEVELOPING WITH CURRENT ROLES IN MEDICAL, LOGISTICS AND PROTECTION.
- ❑ ROBOTS HAVE CONS JUST LIKE ANY COMPUTER ENGINE/PROGRAM THAT CAN LEAD TO RISKS, VULNERABILITY AND DEPENDENCY.
- ❑ AND SEEING THAT UTILIZING ROBOTS IS STILL FAIRLY NEW IT RAISES ETHICAL ISSUES ON WHAT RULES SHOULD BE IN PLACE AND TO WHAT EXTENT.

Cited Works and Credit for Images

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[6] "M160 robotic mine flail," USAASC. [Online]. Available: <https://asc.army.mil/web/portfolio-item/cs-css-m160-robotic-mine-flail/>

[7] "Titan Unmanned Ground Vehicle (UGV)," *Army Technology*, Jun. 2020. [Online]. Available: <https://www.army-technology.com/projects/titan-unmanned-ground-vehicle-ugv/>

[8] <https://www.designweek.co.uk/issues/may-2013/confused-com-unveils-new-branding-and-brian-the-robot/>

[9] Hannah Muphy, "Ethical Hackers 'Hit the Jackpot' as tech groups pay for protection", March 2022, [Online]

[10] Robot and Human Picture, <https://www.freepik.com/free-vector/illustration-robot-vector-graphic> [ON

Additional Questions?

- ❑ Are robots a solution?
- ❑ Will the use of robots in war lead to larger consequences?